

List of Publications - Chen Hajaj - January 2026

REFEREED
JOURNAL
PUBLICATIONS

1. Ron Hirshprung, Ori Lesman, **Chen Hajaj**. The Sound of Emotions, An Artificial Intelligence Approach to Predicting Emotions from Musical Selections. In *Multimedia Systems*. 2026
2. Anat Goldstein, **Chen Hajaj**, Pierre Singer, Orit Raphaeli. Survival Forest Models for ICU Mortality Prediction based on Nutrition and Clinical Factors. In *IEEE Access*. 2026
3. Zafir Avrahami, Moti Zwilling, **Chen Hajaj**. Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions. *IEEE Access*. 2025.
4. Ro'i Lang, Or Haim Anidjar, Sahar Slonimsky, **Chen Hajaj**, Oz Golan, Carmel Matias, Alex Diskin, Evgeny Strokin, Mor Mega. Measuring and Analyzing Defects of Additive Manufactured Ti-6Al-4V Specimens Through Image Segmentation. *Fatigue & Fracture of Engineering Materials & Structures*. 2025.
5. Udi Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. A classification-by-retrieval framework for few-shot anomaly detection to detect API injection. In *Computers & Security*. 2025
6. Or Haim Anidjar, Revital Marbel, Ran Dubin, Amit Dvir, **Chen Hajaj**. Extending Limited Datasets with GAN-Like Self-Supervision for SMS Spam Detection. In *Computers & Security*. 2024
7. Assaf Frajman, Shimon Izhakian, Ori Mekiten, Ori Hadar, Ariel Lichtenstadt, **Chen Hajaj**, Shon Schori, Moshe Heching, Dror Rosengarten and Mordechai R. Kramer. Phenotypical Characteristics of Nontuberculous Mycobacterial Infection in Patients with Bronchiectasis. In *Respiratory Research*. 2024
8. **Chen Hajaj**, Porat Aharon, Ran Dubin, and Amit Dvir. The Art of Time-Bending: Data Augmentation and Early Prediction for Efficient Traffic Classification. In *Expert Systems with Applications*. 2024
9. Anat Goldstein and **Chen Hajaj**. Measuring flight-destination similarity: A multidimensional approach. In *Expert Systems with Applications*. 2024
10. Ofek Bader, Adi Lichy, Amit Dvir, Ran Dubin, and **Chen Hajaj**. OSF-EIMTC: An Open-Source Framework for Standardized Encrypted Internet Traffic Classification. In *Computers Communications*. 2024
11. Orit Raphaeli, Liran Statlender, **Chen Hajaj**, Itai Bendavid, Anat Goldstein, Eyal Robinson, and Pierre Singer. Using Machine-Learning to Assess the Prognostic Value of Early Enteral Feeding Intolerance in Critically Ill Patients: A Retrospective Study. In *Nutrients*. 2023
12. Harel Berger, Amit Dvir, Enrico Mariconti, and **Chen Hajaj**. Breaking the structure of MaMaDroid. In *Expert Systems with Applications*. 2023

13. Ofir Joseph, Avshalom Elmalech, and **Chen hajaj**. Detecting Parallel Covert Data Transmission Channels in Video. In *Electronics*. 2023
14. Or Haim Anidjar, Yannick Estève, **Chen Hajaj**, Amit Dvir, Itshak Lapidot. Speech and Multilingual Natural Language Framework for Speaker Change Detection and Diarization. In *Expert Systems With Applications*. 2023
15. Adi Lichy, Ofek Bader, Ran Dubin, Amit Dvir, **Chen Hajaj**. When a RF Beats a CNN and GRU, Together - A Comparison of Deep Learning and Classical Machine Learning Approaches for Encrypted Malware Traffic Classification. In *Computers & Security*. 2023
16. Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich, Corine Berger. The Online temporary learning groups in higher education: Interactions, compensation, and maximization of achievements in an Israeli case study. In *Journal of Education Culture and Society*. 2023
17. Rotem Bar and **Chen Hajaj**. SimCSE for Encrypted Traffic Detection and Zero-Day Attack Detection. In *IEEE ACCESS*. 2022
18. **Chen Hajaj**, Nitay Hason, and Amit Dvir. Less is More: Robust and Novel Features for Malicious Domain Detection. In *Electronics*. 2022
19. Anat Goldstein and **Chen Hajaj**. The Hidden Conversion Funnel of Mobile vs. Desktop Consumers. In *Electronic Commerce Research and Applications (ECRA)*, 2022
20. Corine Berger, Uzi Ben-Shalom, **Chen Hajaj**, Nitza Davidovich. Patterns of Activity in Temporary Digital Learning Groups, and Its Correlation With Academic Achievement Among University Students in an Israeli University. In *Social Issues in Israel*. 2022
21. Harel Berger, **Chen Hajaj**, and Amit Dvir. Crystal Ball: From Innovative Attacks to Attack Effectiveness Classifier. *IEEE ACCESS*. 2021
22. Ron Hirschprung and **Chen Hajaj**. Prediction Model for the Spread of the COVID-19 Outbreak in the Global Environment. *Heliyon*. 2021
23. Or Anidjar, Itshak Lapidot, **Chen Hajaj**, Amit Dvir and Issachar Gilad. Hybrid Speech and Text Analysis Methods for Speaker Change Detection. In *IEEE/ACM Transactions on Audio Speech and Language Processing*. 2021
24. **Chen Hajaj**, Zlatko Joveski, Sixie Yu, Yevgeniy Vorobeychik. Robust Coordination in Adversarial Social Networks: From Human Behavior to Agent-Based Modeling. in *Network Science*, 2021.
25. Amit Dvir, Angelos K. Marnierides, Ran Dubin, Nehor Golan, and **Chen Hajaj**. Encrypted Video Traffic Clustering Demystified. In *Computers & Security*, 96 ,2020.

26. **Chen Hajaj** and David Sarne. Selective Opportunity Disclosure at the Service of Strategic Information Platforms. In *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(5):1133–1164, 2017.
27. **Chen Hajaj**, Noam Hazon, and David Sarne. Enhancing Comparison Shopping Agents through Ordering and Gradual Information Disclosure. In *Autonomous Agents and Multi-Agent Systems (JAAMAS)*, 31(3):696–714, 2017.
28. **Chen Hajaj**, Noam Hazon, and David Sarne. Improving Comparison Shopping Agents’ Competence through Selective Price Disclosure. In *Electronic Commerce Research, and Applications (ECRA)*, 14(6) 563–581, 2015.

REFEREED
CONFERENCE
PUBLICATIONS

1. Meir Litman and **Chen Hajaj**. Cleaner Adversarial CAPTCHAs: Intelligent Targets and Precise Noise for Usable Security. In *The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2026
2. Revital Marbel, Yanir Cohen, Ran Dubin, Amit Dvir and **Chen Hajaj**. Cloudy with a Chance of Anomalies: Dynamic Graph Neural Network for Early Detection of Cloud Services’ User Anomalies. In *the 34th International Conference on Computer Communications and Networks (ICCCN)*. 2025
3. Simona Lisker, Ayelet Botman, **Chen Hajaj**, Ran Dubin, Amit Dvir. Optimized File Type Detection and One-Shot Retrieval. In *the IEEE International Conference on Communications (ICC)*. 2025
4. Zohar Simhon, Matan Weiss, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. A New D-MAGIC: Dynamic Model for Cybersecurity Attack Detection Using GNNs into Clustering. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025
5. Yehonatan Zion, Porat Aharon, Ran Dubin, Amit Dvir, **Chen Hajaj**. Enhancing Encrypted Internet Traffic Classification Through Advanced Data Augmentation Techniques. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025
6. Angelos Marnierides, **Chen Hajaj**, Revital Marbel, Ran Dubin, Amit Dvir. PQClass: Classification of Post-Quantum Encryption Applications in Internet Traffic. In *Proceedings of the IEEE International Conference on Communications (ICC)*. 2025.
7. Anat Goldstein and **Chen Hajaj**. Warm Recommendation: Enhancing Cold Start Recommendations Using Multimodal Product Representations. In *International conference of the Association for Information Systems (ICIS)*. 2024
8. Nathan Dillbary, Roi Yozevitch, Amit Dvir, Ran Dubin, and **Chen Hajaj**. In *Hidden in Time, Revealed in Frequency: Spectral Features and Multiresolution Analysis for Encrypted Internet Traffic Classification*. In *the Proceedings of IEEE Consumer Communications & Networking Conference*. 2024

9. Anat Goldstein and **Chen Hajaj**. Warming Up the Cold Start: A Multimodal Approach. In *Special Interest Group on Decision Support and Analytics (SIGDSA) Symposium*. 2023
10. Orit Raphaeli, Liran Statlander, **Chen Hajaj**, Itai Ben David, and Pierre Singer. Protein Intake and Clinical Outcomes of Critically Ill Patients. In *the 44th European Congress for Clinical Nutrition and Metabolism (ESPEN)*. 2022
11. Anat Goldstein and **Chen Hajaj**. Measuring Product Similarity: A Multidimensional Approach. In *the Eighteenth Symposium on Statistical Challenges in Electronic Commerce Research*. 2022
12. Ofek Bader, Adi Lichy, **Chen Hajaj**, Ran Dubin, and Amit Dvir. MalDIST: From Encrypted Traffic Classification to Malware Traffic Detection and Classification. In *the Proceedings of IEEE Consumer Communications & Networking Conference*. 2022
13. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer, and Uri Gabbay. Using the Cardio-Vascular Index (CVRI) To Predict Mortality in Septic Shock. In *the 40th International Symposium on Intensive Care & Emergency Medicine*. 2022
14. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Support Early Prediction of Feeding Intolerance in Critically Ill Patients. In *Intensive Care Medicine Experimental*. 2021
15. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Using Machine Learning to Compare Gastric Residual Volume Thresholds as Predictors of Clinical Outcomes in Critically Ill Patients. In *International Symposium of Intensive Care and Emergency Medicine*. 2021
16. Or Anidjar, Itshak Lapidot, **Chen Hajaj** and Amit Dvir. A Thousand Words are Worth More Than One Recording: Word-Embedding Based Speaker Change Detection. In *Interspeech*. 2021
17. Orit Raphaeli, **Chen Hajaj**, Itai Ben David, Anat Goldstein, Elad Chen, Pierre Singer. Feeding Intolerance as a Predictor of Clinical Outcomes in Critically Ill Patients: A Machine Learning Approach. In *the 43rd European Society for Clinical Nutrition and Metabolism (ESPEN)*. 2021
18. Uzi Ben Shalom, **Chen Hajaj**, Nitza Davidovich and Corinne Berger. Online temporary learning groups in higher education – Interactions, compensation, and maximization of achievements in an Israeli case-study. In *the 1st international e-conference on teacher education generation next: perspectives, opportunities & challenges*. 2021
19. Vitali Demianiuk, **Chen Hajaj**, and Kiril Kogan. PCL: Packet Classification with Limited Knowledge. *INFOCOM*. 2021

20. Anat Goldstein and **Chen Hajaj**. The Different Path to Purchase of Mobile and Desktop Consumers: Analyzing Consumers' Progress in the Conversion Funnel Using Hidden Markov Models. In *Statistical Conference in E-Commerce Research (SCECR)*. 2020
21. Nitay Hason, Amit Dvir, and **Chen Hajaj**. Robust Malicious URL Detection. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
22. Liang Tong, Bo Li, **Chen Hajaj**, Chaowei Xiao, Ning Zhang, Yevgeniy Vorobeychik. Improving Robustness of ML Classifiers against Realizable, Evasion Attacks Using Conserved Feature. In the *28th USENIX Security Symposium*. 2019
23. Yevgeniy Vorobeychik, Myrna Wooders, Greg Leo, Matthew Chambers, **Chen Hajaj**, Jian Lou, and Martin Van der Linden. Non-Cooperative Team Formation and a Team Formation Mechanism. In the *11th annual conference of the Israeli chapter of the game theory society*. 2019
24. **Chen Hajaj**, Sixie Yu, Zlatko Joveski, Yifan Guo, and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *the International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2019
25. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin, and Daniel Fabbri. Crowdsourcing Clinical Chart Reviews. In *AMIA Joint Summits on Translational Science*. 2018
26. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial task assignment. In *International Joint Conference on Artificial Intelligence and European Conference on Artificial Intelligence (IJCAI-ECAI)*. 2018
27. Cheng Ye, Joseph Coco, Anna Epishova, **Chen Hajaj**, Henry Bogardus, Laurie Novak, Joshua Denny, Yevgeniy Vorobeychik, Thomas Lasko, Bradley Malin and Daniel Fabbri. A Crowdsourcing Framework for Medical Data Sets. In *American Medical Informatics Association (AMIA)*. 2018
28. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Enhancing Crowdworkers' Vigilance. In *International Joint Conference on Artificial Intelligence (IJCAI)*. 2017
29. Avshalom Elmalech, David Sarne, Esther David, and **Chen Hajaj**. Extending Workers' Attention Span Through Dummy Sub-Goals. In *AAAI Conference on Human Computation and Crowdsourcing (HCOMP)*. 2016. **Best paper finalist.**
30. **Chen Hajaj**, John P. Dickerson, Avinatan Hassidim, Tuomas Sandholm and David Sarne. Strategy-Proof and Efficient Kidney Exchange Using a Credit Mechanism. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2015

31. Tal Darom, Yaniv Gur, **Chen Hajaj** and Yosi Keller. Automatic Anatomical Shape Correspondence and Alignment Using Mesh Features. In *International Symposium on Biomedical Imaging (ISBI)*. 2015
32. **Chen Hajaj** and David Sarne. Strategic Information Platforms - Selective Disclosure and The Price of "Free". In *ACM Conference on Economics and Computation (EC)*. 2014
33. **Chen Hajaj**, Noam Hazon, and David Sarne. Ordering Effects and Belief Adjustment in the Use of Comparison Shopping Agents. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2014
34. **Chen Hajaj**, Noam Hazon, David Sarne, and Avshalom Elmalech. Search More, Disclose Less. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2013

SHORT PAPERS
AND REFEREED
WORKSHOPS

1. Brigitta R Schvarcz and **Chen Hajaj**. Exploring Student Perspectives of the Impact of VR in ESP. In *TESOL*. 2024
2. Bar Vinder, **Chen Hajaj**, and Idit Avrahami. A method for evaluating the performance of a novel percutaneous Left Ventricular Assist Device and optimizing its design. In *VPH 2022 PHYSIOLOGICAL HUMAN CONFERENCE*. 2022
3. Harel Berger, **Chen Hajaj**, and Amit Dvir. Evasion is not enough: A case study of Android malware. In the *4th International Symposium on Cyber Security Cryptology and Machine Learning (CSCML)*. 2020
4. **Chen Hajaj** and Yevgeniy Vorobeychik. Adversarial Randomized Task Assignment, 6th Day on Computational Game Theory. 2019
5. **Chen Hajaj**, Sixie Yu, Zlatko Joveski and Yevgeniy Vorobeychik. Adversarial Coordination on Social Networks. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence(AI³)*. 2018
6. Jian Lou, **Chen Hajaj**, Myrna Wooders and Yevgeniy Vorobeychik. Automated Mechanism Design for the Roommates Problem. In *The AAMAS-IJCAI workshop on agents and incentives in artificial intelligence (AI³)*. 2018
7. Jian Lou, Martin Van der Linden, Pranav Batra, **Chen Hajaj**, Greg Leo, Yevgeniy Vorobeychik and Myrna Wooders. Rotating Proposer Mechanisms for Team Formation. In *Workshop on Cooperative Games in Multiagent Systems (CoopMAS)*.2017. **Most visionary paper.**
8. **Chen Hajaj**, David Sarne, and Lea Perets. Advanced Service Schemes for a Self-Interested Information Platform. In *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2014

BOOK CHAPTERS

Sofia Ceppi, Esther David, **Chen Hajaj**, Valentin Robu, and Ioannis A. Vetsikas, editors, Agent-Mediated Electronic Commerce: Designing Trading Strategies and Mechanisms for Electronic Markets, Springer, LNBIP 271, 2017.